PURDUE UNIVERSITY | Elmore Family School of Electrical and Computer Engineering

ECE 20002 – Electrical Engineering Fundamentals II
Summer 2026 – Syllabus

1. Course Information

- ECE 20002 – Electrical Engineering Fundamentals II – 3 Credit Hours
- Instructor: Dr. Phil Walter
- Prerequisites: ECE 20001 (minimum grade of C)
- Section 002 – CRN 12559 – Summer 2026 (202630) – June 15 to August 7
  - Lecture MTWRF – 08:40 – 9:40am – Synchronous Online*
- Section 003 – CRN 38189 – Summer 2026 (202630) – June 15 to August 7
  - Lecture MTWRF – 08:40 – 9:40am – Synchronous Online*

*** Only some exams and office hours will be synchronous, the lectures will be pre-recorded and will not be offered live.**

2. Course Description

Continuation of Electrical Engineering Fundamentals I. The course addresses mathematical and computational foundations of circuit analysis (differential equations, Laplace Transform techniques) with a focus on application to linear circuits having variable behavior as a function of frequency, with emphasis on filtering. Variable frequency behavior is further considered for applications of electronic components through single-transistor and operational amplifiers.

3. Course Learning Outcomes

The ABET Program Outcome "An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics." is mapped into this course and the following Course Learning Outcomes support this ABET Program Outcome.

A student who successfully fulfills the course requirements will have demonstrated an ability to:

- Analyze and design transistor amplifiers at low, mid and high frequencies,
- Analyze 2nd-order linear circuits with sources and/or passive elements,
- Compute responses of linear circuits with and without initial conditions via one-sided Laplace transform techniques, and
- Compute responses to linear circuits using transfer-function and convolution techniques.

4. Course Staff

Name	Title / Role	Email
Dr. Phil Walter	Instructor	
Boyu Gong	Grad Teaching Assistant	
Jonathan Lane	Grad Teaching Assistant	
Vishnu Nampoothiri	Grad Teaching Assistant	
David Nelson	Grad Teaching Assistant	

5. **Office Hours**
   - **Dr. Walter**
     - Book Me times: 30-minute virtual meetings on MS Teams
       Available: MWF 10:00–12:00pm and TR 1:00-3:00pm
       () →
     - Additional office hours by request.

   - **GTAs**
     TBD – Will update and post in the first week of the course

6. **Learning Resources, Technology, and Textbooks**
   - **Required Textbook**
     - *Electrical Engineering Fundamentals II: Purdue University Lectures from ECE 20002*, Talavage and Terlep, Version 4– Fall 2025. ISBN 978-1098760045, Hardcopy or e-textbook; available through Amazon.com.
   - **Additional Reading Material and Optional Textbooks**
     - GTA Recitations and other material posted in the Weekly Resources sub-modules in Brightspace
     - (Optional Text) *Electric Circuits*, James Nilsson and Susan Reidel, Pearson, 11th Edition, 2018, ISBN 978-0134746968
     - (Optional Text) *Circuits, Signals, and Systems*, W.M. Siebert, MIT Press, 1985 ISBN 978-0262690959– Microelectronic Circuits, Adel Sedra and Kenneth Smith, Oxford University Press, Seventh Edition, ISBN 978-0-19-933914-3
     - (Optional Text) *Microelectronic Circuit Design*, Richard Jaeger and Travis Blalock, McGraw Hill Education, Fifth Edition, ISBN 978-0-07-352960-8
   - **Software/Web and Other Resources**
     - Reliable internet access for this online course
     - A private and quiet place to take the scheduled Viva Voce Exams
     - Brightspace access
     - Piazza
     - Microsoft Teams
     - (Optional) Matlab or similar software
     - (Optional) Circuit simulators such as LTSpice, Falstad, Circuit Lab, Easy EDA, or similar

7. **Course Logistics**
   - **Lectures**
     All lectures will be pre-recorded and the links posted in that week's Brightspace module. Expect that there will be ~5-7 hours of recorded video lectures per week. **There will be no live lectures and no attendance requirements.**
   - **Recitation Sessions**
     The GTAs will be creating some recorded recitation sessions, where they will work through problems similar to that week's homework.
   - **Homework (7)**
     There will be a homework assignment in each of the first 7 weeks of the 8-week course (the last week is devoted to the final exams). All homework will be accessed, submitted, and graded via Brightspace. Any resources are allowed to be used to complete the homework. The homework is worth 10% of your course grade.

- **Online Exams (3)**
  There will be three 60–90-minute Online Exams. These exams must be completed within a single continuous 4-hour session and taken between midnight and 11:59pm Eastern Daylight Time on the scheduled date of the exam. The Online Exams will be accessed, submitted, and graded via Brightspace. The ALLOWED resources for the Online Exams include any materials on the course Brightspace, any materials and posts on the course Piazza, the required Textbook, all your personal coursework, and any course notes that you took during the course. UNAUTHORIZED materials include other web-based materials, LLMs and generative AI, other people, or any other outside help. You will be required to sign a statement at the end of each Online Exam stating that you used no unauthorized material or outside help while taking the exam. Each Online Exam is worth 20% of your course grade (60% for all 3 exams).
- **Online Final Exam (1)**
  There will be one 90–120-minute Online Final Exam. This exam must be completed within a single continuous 4-hour session and taken between midnight and 11:59pm Eastern Daylight Time on the scheduled date of the exam. The Online Final Exam will be accessed, submitted, and graded via Brightspace. The ALLOWED resources for the Online Final Exam include any materials on the course Brightspace, any materials and posts on the course Piazza, the required Textbook, all your personal coursework, and any course notes that you took during the course. UNAUTHORIZED materials include other web-based materials, LLMs and generative AI, other people, or any other outside help. You will be required to sign a statement at the end of the Online Final Exam stating that you used no unauthorized material or outside help while taking the exam. The Final Online Exam is worth 30% of the course grade.
- **Viva Voce Exams (2)**
  There will be 2 Viva Voce Exams, one after Online Exam 2, and one after the Online Final Exam. These Viva Voce Exams will be ~15-minute one-on-one online interview type sessions on Microsoft Teams with either the instructor or one of the GTAs. These Viva Voce Exams sessions will be scheduled for each student within a 30-minute Microsoft Teams session (to allow for setup, accommodations, or other issues). **We will schedule these Teams meetings via MS Booking at least a week ahead of the scheduled Viva Voce Exams**. You must download and test Microsoft Teams on at least one device (laptop or computer is recommended, but it will work on a tablet or phone) well before the first scheduled Viva Voce Exam. **The only ALLOWED resources for the Viva Voce Exams are the student's memory and any material that the interviewer supplies during the session** (via sharing their screen). **The Viva Voce Exam sessions will be recorded.** At the beginning of the session, the student will be asked to scan the camera around the room to ensure that there are no other devices or people in the immediate vicinity (**you must have a private and quiet place available for these scheduled Viva Voce Exams**). **Then the student will be required to show a photo ID** (Purdue Student ID, Drivers License, Passport, etc.). The interviewer will then share their screen, and the student should make any adjustments so that they can read the words/equations on the screen clearly with no further adjustments needed on the student's device. During the Viva Voce Exam interview, there should be no need for the student to type, use a mouse or other pointing device, or move the camera. We only want to see the student and hear the student speak and we want the student to only watch their screen or the camera while answering the interviewer's questions. To ensure this, we want the camara positioned such that the student's face and both hands are always visible at all times during the interview. After this the Viva Voce Exam interview will start. The interviewer will show two (or possibly 3) problems where each problem is one that the student had correctly answered from their course exams or homework. For each problem the student will be asked to explain how to solve the problem. The interviewer may write or show the steps as the student explains and/or show

additional equations. The student must be able to clearly demonstrate to the interviewer's satisfaction that the student knows how to solve that problem. If the student can successfully describe how to solve the first problem, then they will be shown a second problem. If the student can successfully describe how to solve the second problem, then they will have successfully passed the Viva Voce Exam. However, if the student has difficulty or cannot properly explain how to solve a problem, they will be given one additional chance with a third problem. **To pass the Viva Voce Exam, a student must be able to clearly demonstrate to the interviewer's satisfaction that the student knows how to solve two problems, where each problem is one that the student had correctly answered from one of their course exams or homework.** The Viva Voce Exams are pass/fail only. **If a student fails both Viva Voce Exams, they will automatically fail the course. If a student fails only one of their Viva Voce Exams, they will have a reduction of 15/100 points on their course grade. If a student passes both Viva Voce Exams, then their earned course grade stands.**

8. Course Communication

- **The first point of contact for any questions about this course should be the course Piazza**, except for any highly personal communications such as accommodation requests. The advantage of Piazza is that it may be faster to receive an answer rather than waiting for an open office hour or an email reply, and it creates a searchable repository of questions/answers that is available to all students anytime. With almost 200 students enrolled in the course it is very unlikely that no other student has the same question. You may also submit questions anonymously (anonymous to your classmates, not to the course staff, sorry).
- **The second point of contact for any questions related to the course materials are the GTA Office Hours.** Please try Piazza first, then the GTAs, and then Dr. Walter for course material questions.
- **The second point of content for any questions related to course administration is Dr. Walter's Office Hours.** Please try Piazza first and then communicate directly to Dr. Walter in his office hours or via email for any questions on the course administration or grading.
- **The point of contact for highly personal issues should be Dr. Walter's office hours or an email.**

9. Assessments and Course Grade

Assessment	Date	Allotted Time	Format	Course Grade
Homework	Weekly	1 Week	via Brightspace	10%
Online Exam 1	July 2	60-90-min within 4-hours	via Brightspace	20%
Online Exam 2	July 17	60-90-min within 4-hours	via Brightspace	20%
Viva Voce Exam 1	July 18-22	15-min within 30-min mtg	MS Teams	P/F
Online Exam 3	July 29	60-90-min within 4-hours	via Brightspace	20%
Online Final Exam	August 3	90-120-min within 4-hours	via Brightspace	30%
Viva Voce Exam 2	Aug 4-7	15-min within 30-min mtg	MS Teams	P/F
			Total	100%

Your grade for this course is intended to assess how well you have met the course learning outcomes listed above in Section 3. I do not curve the final Course Grade. If it is determined that there was some mistake or unfairness in an Assessment, I may curve the grade on that Assessment to compensate. At the end of the semester, final Course Grades will be calculated based on the grades received on each of the Assessments and adjusted based on any failures in the Viva Voce

Exams. The final numerical Course Grade will be mapped into final letter grades using the following table. Fractional elements within 0.5 will be rounded up to the next letter grade.

A+:	97 – 100%	B+:	87 – 89%	C+:	77 – 79%	D+:	67 – 69%		
A:	93 – 96%	B:	83 – 86%	C:	73 – 76%	D:	63 – 66%		
A-:	90 – 92%	B-:	80 – 82%	C-:	70 – 72%	D-:	60 – 62%	F:	Below 60%

Note: Regardless of the final Course Grades described above, a student must pass at least one of the Viva Voce Exams to pass the course. Fail one of the Viva Voce Exams: Receive a 15/100 point reduction of the Course Grade. Fail both Viva Voce Exams: Fail the course.

10. Attendance and Late Submissions

This is a fully online course and with the exception of the Viva Voce Exams and available office hours, is completely asynchronous with no attendance requirements. **The Online Exams and Homework Assessments must be submitted before the posted deadlines, which will always be before 11:59pm Eastern Daylight Time on the scheduled date. The Brightspace submission will close after the deadline and only with an approved excused absence that follows the Purdue Absence Policy will late submissions or rescheduled due dates be accepted or accommodated. Notify me (xxxxxxxxxxxxxxx) beforehand (if at all possible) and as early as possible of any absences or for accommodations on deadlines.** Remember that you have at least a week for the Homework and a 24-hour window to take the Online Exams. **Missing an Assessment deadline or missing a scheduled Viva Voce Exam without an excused absence accommodation will result in a 0 or failure for that Assessment.**

11. Course Schedule

Below is a calendar showing the Homework due dates and dates of the Exams. A more detailed calendar with lecture topics follows.

ECE 20002 - Summer 2026	Electrical Engineering Fundamentals II			Tenative Schedule - Subject to Change	
	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
	June 15, 2026	June 16, 2026	June 17, 2026	June 18, 2026	June 19, 2026	June 20, 2026
Week 1	Classes Begin				Homework 1 Due	
	June 22, 2026	June 23, 2026	June 24, 2026	June 25, 2026	June 26, 2026	June 27, 2026
Week 2					Homework 2 Due	
	June 29, 2026	June 30, 2026	July 1, 2026	July 2, 2026	July 3, 2026	July 4, 2026
Week 3			Homework 3 Due	Online Exam 1	Holiday No Classes	Holiday No Classes
	July 6, 2026	July 7, 2026	July 8, 2026	July 9, 2026	July 10, 2026	July 11, 2026
Week 4					Homework 4 Due	
	July 13, 2026	July 14, 2026	July 15, 2026	July 16, 2026	July 17, 2026	July 18, 2026
Week 5				Homework 5 Due	Online Exam 2	Viva Voce Exams
	July 20, 2026	July 21, 2026	July 22, 2026	July 23, 2026	July 24, 2026	July 25, 2026
Week 6	Viva Voce Exams	Viva Voce Exams	Viva Voce Exams		Homework 6 Due	

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	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Week 7	July 27, 2026	July 28, 2026 Homework 7 Due	July 29, 2026 Online Exam 3	July 30, 2026	July 31, 2026	August 1, 2026
Week 8	August 3, 2026 Online Final Exam	August 4, 2026 Classes End Viva Voce Exams	August 5, 2026 Viva Voce Exams	August 6, 2026 Viva Voce Exams	August 7, 2026 Summer Session Ends Viva Voce Exams	August 8, 2026

Week/date	Lecture Topics	Textbook Chapters	Video Lectures
Week 1 June15-20	- Introduction/Syllabus/Calendar - Field-Effect Transistors - MOSFET Operational Concepts - MOSFETs in Circuits; DC Operational Concepts - Transistors as Amplifiers **- Homework 1 Deadline (before 11:59pm on June 19)**	 2 2 3 4	0 (???) 1 (1:20:43) 2 (16:47) 3 (42:09) 4 (35:50)
Week 2 June 22-27	- Common-Source Amplifier Circuits - Single-Transistor Amplifier Circuits - Freq. Range for FET Amplifier Circuits (Low Freq. Cutoff) - Freq. Range for FET Amplifier Circuits (High Freq. Cutoff) **- Homework 2 Deadline (before 11:59pm on June 26)**	5 6 30 31	Will update and post
Week 3 June 29 – July 2	- Op Amp Basics - Op Amp Circuits - Op Amp Circuits (cont.) **- Homework 3 Deadline (before 11:59pm on July 1)** **- Online Exam 1 (July 2 – submit before 11:59pm)**	1,7 1,7 1,7 - -	Will update and post
Week 4 July 6-11	- ODEs for RC and RL Circuits - ODEs for LC Circuits - ODEs for RLC Circuits - Analysis of Switched Circuits **- Homework 4 Deadline (before 11:59pm on July 10)**	8,9 10 11 12 -	Will update and post
Week 5 July 13-18	- Convolution I - Convolution II - Impulse Response **- Homework 5 Deadline (before 11:59pm on July 16)** **- Online Exam 2 (July 17 – submit before 11:59pm)** **- Viva Voce Exam 1 Sessions (July 18-22)**	13,14 13,14 16 - - -	Will update and post
Week 6 July 20-25	- Introduction to Laplace Transforms - Properties of Laplace Transforms - Inverse Laplace Transforms - Impedance, Admittance, and Zero-State Response - Initial Conditions in Laplace Transforms - Decomposition of the Complete Response - Transfer Functions - The Complex Plane and Stability **- Homework 6 Deadline (before 11:59pm on July 24)**	18 19 20 21 22 24 23 25 -	Will update and post

Week 7 July 27- August 1	- Resonance and 2nd-Order Systems - Basics of Frequency Response I - Basics of Frequency Response II - Magnitude and Frequency Scaling - Bandpass Response of 2nd-Order Systems **- Homework 7 Deadline (before 11:59pm on July 28)** **- Online Exam 3 (July 29 – submit before 11:59pm)** - Passive Low-Pass Filter - Butterworth Low-Pass Filter Design - Butterworth High-Pass Filter Design from LPF - ActiveLow-PassFilters	26 27 27 28 29 - - 32 33 34 35	Will update and post
Week 8 August 3-7	- **Online Final Exam (August 3 – submit before 11:59pm)** - **Viva Voce Exam 2 Sessions (August 4-7)**	- -	- -

12. Academic Integrity, Copyright, AI, and Professionalism

Any student found using unauthorized material or outside help on any of the Oline Exams or Viva Voce Exams will fail the course and be reported to the dean.

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information is submitted the greater the opportunity for the university to investigate the concern. More details are available on the Academic Resources table on your Brightspace homepage.

See the University Policies and Statements section of Brightspace for guidance on Use of Copyrighted Materials. Effective learning environments provide opportunities for students to reflect, explore new ideas, post opinions openly, and have the freedom to change those opinions over time. Students and instructors are the authors of the works they create in the learning environment. As authors, they own the copyright in their works subject only to the university's right to use those works for educational purposes. **Students may not copy, reproduce, or post to any other outlet (e.g., YouTube, Facebook, or other open media sources or websites) any work in which they are not the sole or joint author or have not obtained the permission of the author(s). I.E., Do not post any of the course material anywhere!**

Unless otherwise noted, students are expected to do their own work, and not copy the work of any other individual, past or present. Any and all sources used in the completion of work for this course should be properly referenced, and where appropriate, the level of original work performed by the student should be noted. Modern Large Language Models/AI are wonderful tools that can help us do our work better and faster. However, no matter how much effort you spend in developing the keywords and prompts for these tools, the resulting output is not your own work. Therefore, if these AI tools are used in your work, you must properly cite the tools that you used and include your keywords and prompts. Otherwise, using the output of these tools is plagiarism and is academic dishonesty.

Any documented case of academic dishonesty will result in a failing grade for the course as well as possible disciplinary action. All cases of academic dishonesty will be reported to the ECE

Associate Head as well as to the Dean of Students. **A professional does not take credit for the work of somebody else.**

13. Nondiscrimination Statement

Purdue University is committed to maintaining a community that recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her potential. A hyperlink to Purdue's full Nondiscrimination Policy Statement is included in the Academic Resources table on your Brightspace homepage.

14. Accessibility

Every member of our course should be able to access, use, and learn from the materials we share. This includes all course-related digital content that you and I share in the course. This approach helps promote equal access for everyone at Purdue and is mandated federally by Title II of the Americans with Disabilities Act (ADA). We will work together to provide this access within our Brightspace course.

- My part, as instructor, is to make sure all course materials shared to Brightspace, such as documents, slides, videos and audio, and images, meet accessibility guidelines and to assist you in making sure anything you share is accessible.
- Your role, as a student, is to make sure anything you post for other students to engage with is also accessible, such as peer grading, peer feedback, and discussion board posts. This expectation is built into all course assignments that require you to post to Brightspace.
- A good starting place for you is to bookmark and review the Innovative Learning Accessibility Checklist for guidance on creating accessible materials.
- When selecting materials to share on our Brightspace from Purdue Libraries catalog or databases, best practices include choosing items that:
  - Can be downloaded in full
  - Are available in EPUB or HTML formats
  - Include alternative text for written materials or captions for audio/visual content

Purdue University strives to make learning experiences accessible to all participants. If you anticipate or experience physical or academic barriers based on disability, you are encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247, as soon as possible.

If the Disability Resource Center (DRC) has determined reasonable accommodations that you would like to utilize in my class, you must release your Course Accommodation Letter to me. Instructions on sharing your Course Accommodation Letter can be found by visiting: How To Use Your Course Accommodation Letter. Additionally, you are strongly encouraged to contact me as soon as possible to discuss implementation of your accommodation.

15. Mental Health/Wellness Statement

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try Therapy Assistance Online (TAO), a web and app-based mental health resource available courtesy of Purdue Counseling and Psychological Services (CAPS)**.** TAO is available to all students at any time by creating an account on the TAO Connect website, or downloading the app from the App Store or Google Play. It offers free, confidential well-being resources through a self-guided program informed by psychotherapy research and strategies that may aid in overcoming anxiety, depression and other concerns. It provides accessible and effective resources including short videos, brief exercises, and self-reflection tools.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 a.m.- 5 p.m.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions in West Lafayette with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is free and can be done on BoilerConnect. Students in Indianapolis will find support services curated on the Vice Provost for Student Life website.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS offices in West Lafayette or Indianapolis.

16. Emergency Preparation

In the event of a major campus emergency, course requirements, deadlines, and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted on the course's Brightspace page. In any case, you are expected to read your @purdue e-mail on a frequent basis.

17. Basic Needs Program

If you are facing challenges securing basic needs such as food, housing, transportation, health services, or access to technology or childcare resources and believe this may affect your performance in the course, please contact the Office of the Dean of Students (ODOS) to help coordinate with community resources. These services vary by location. In **West Lafayette**, see the Basic Needs Program website, or email basicneeds@purdue.edu. To connect with a Student Support Generalist on the **Indianapolis** campus, contact them by phone at 765-495-7797 or email studentlifeindy@purdue.edu.